FLIGHT BOOKING SYSTEM USING CIRCULAR LINKED LIST

1. Problem Statement

The Flight Booking System is a C++ application developed using a Circular Linked List to manage flight booking records. It allows users to book flights, display all bookings, search for a booking using Booking ID, and cancel bookings. The program also demonstrates dynamic memory allocation functions such as `malloc()`, `calloc()`, `realloc()`, and `free()`.

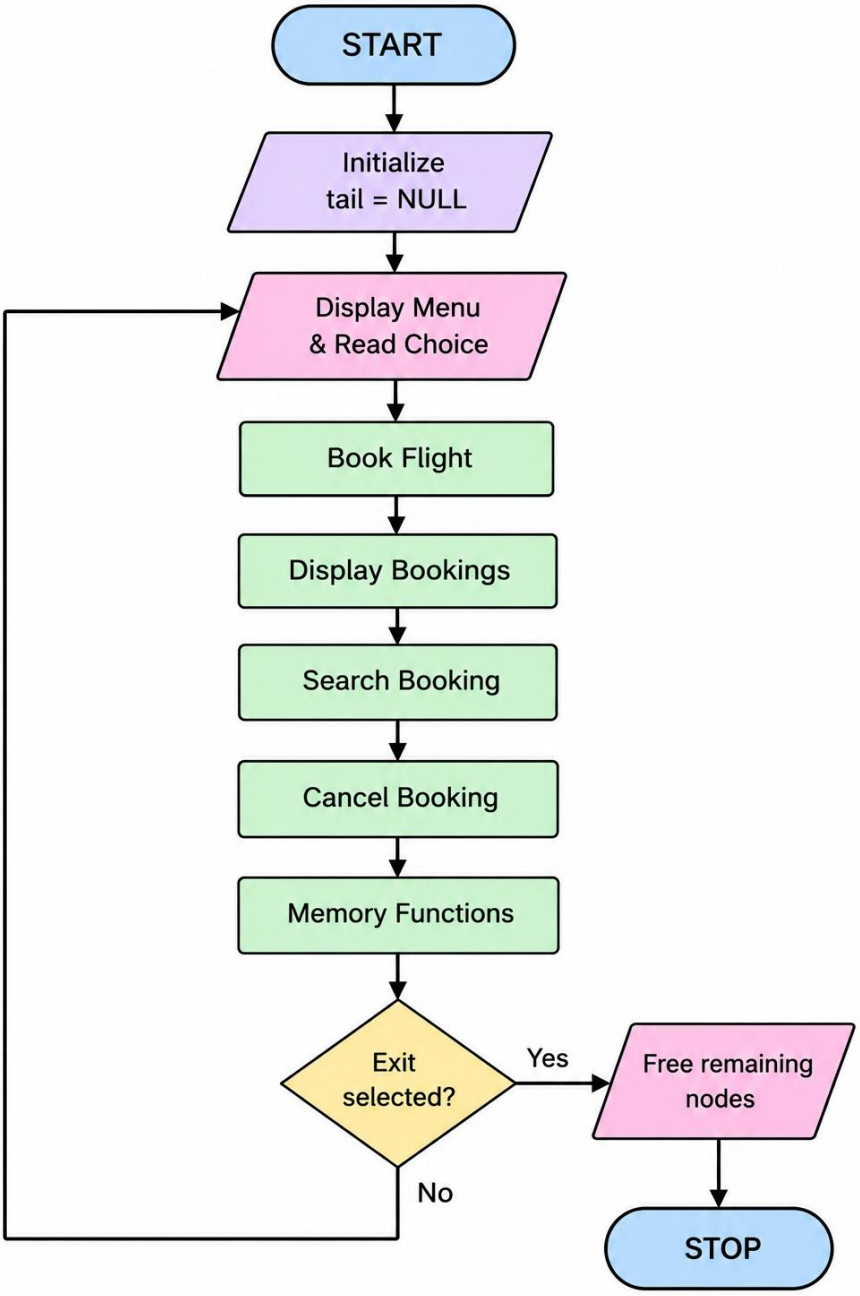
2. Objectives

- To implement a Circular Linked List in C++.
- To insert and display flight booking records.
- To search bookings using Booking ID.
- To cancel a booking by deleting its node.
- To understand dynamic memory allocation and deallocation.
- To demonstrate circular traversal of linked-list nodes.

3. Algorithm

1. Start the program.
2. Initialize `tail = NULL`.
3. Display the menu and accept the user's choice.
4. Create a node and insert flight details to book a flight.
5. Display all bookings by traversing the Circular Linked List.
6. Search for a booking using its Booking ID.
7. Cancel a booking by updating links and deleting the node.
8. Demonstrate `malloc()`, `calloc()`, `realloc()`, and `free()`.
9. Repeat the menu until the user selects Exit.
10. Release all remaining nodes and stop the program.

4. Flowchart



5. Conclusion

The Flight Booking System was successfully implemented using a Circular Linked List in C++. It performs booking insertion, display, searching, and cancellation. The project demonstrates circular traversal and the use of dynamic memory allocation functions, helping to understand linked-list operations and memory management.